

SANKALP SHANKAR

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Objective

ML Engineer who has shipped LLM-powered web applications end-to-end — from RAG pipelines and vector retrieval to FastAPI backends and React frontends. I take ownership of full project lifecycles, write production-ready code, and care deeply about the problems I work on.

Education

Acharya Institute of Technology

Bachelor of Engineering in AIML

- SGPA: 8.7/10.0

Expected July 2027

Bangalore, India

Technical Skills

Languages & Frameworks: Python, JavaScript, HTML5, CSS, React, Next.js, FastAPI, Express.js, Node.js

AI / LLM: OpenAI, Gemini, LangChain, RAG, Vector Databases, TensorFlow, PyTorch, XGBoost, Scikit-learn, NLTK

Data & Databases: NumPy, Pandas, SQL, MongoDB, Redis

Cloud & DevOps: AWS, Docker, CI/CD, Git, GitHub, Jupyter Notebook, VS Code

Projects

PharmaGuard | *LLM-Enabled Clinical Decision Support System*

2026

- Built end-to-end FastAPI + React platform analyzing genomic VCF files to generate CPIC-guided prescribing recommendations
- Integrated PharmCAT pipeline with deterministic risk engine and Llama 3.3 (via Groq) to produce clinician-ready explanations; exported EHR-ready JSON for downstream system integration
- Containerized all services with Docker and configured structured APIs for production deployment on AWS

SwitchBoard | *LLM Gateway & Multi-Provider Routing Layer*

2026

- Built an abstraction layer routing requests across OpenAI and Gemini/Groq with automatic failover, retries, and quota monitoring
- Implemented model fallback chains and centralized logging, increasing reliability of deployed LLM-based web applications
- Enabled seamless provider switching without any changes to application-layer code, reducing integration overhead

MediAssist AI | *RAG-Based Agentic CDSS — Hackathon Runner-up*

2025

- Built a RAG pipeline using LangChain and a vector database for structured medical knowledge retrieval with OpenAI-backed agents
- Integrated XGBoost predictive models for explainable differential diagnosis with confidence-scored outputs
- Deployed full-stack application with an automated NLP feedback loop for continuous output improvement

Red Agent | *Autonomous Web Scanning Platform*

2026

- Built FastAPI-based scanning engine with orchestrated job lifecycle and Redis-backed state management, deployed on AWS with CI/CD
- Generated structured vulnerability reports with severity scoring and exportable API responses
- Designed modular architecture enabling extensible exploit modules and scalable scan execution

Achievements & Leadership

Technical Achievements

2022 – Present

Competitive Programming, Hackathons & National Tech Events

Multiple Platforms

- 7× Competition Winner: 4 Hackathon wins (1st Place – State Level, Runner-up – 48hr Hackathon, 3rd Place – Zonal) and 2 National AI Ideathon titles
- Overall Champions – XActitude Tech Fest (Kristu Jayanti College); Gold – Tech Quiz & CTF, Silver – SQL Competition, 3rd – Code Relay (HackerRank) & Prompt Engineering
- Maintained ~1600 LeetCode rating with 260+ problems solved across Data Structures, Algorithms, and System Design fundamentals

Event Head – Tech Habba (3-Day Technical Festival)

2024

Acharya Institute of Technology

Bangalore, India

- Led and oversaw all departmental committees for a 3-day institute-wide technical festival
- Managed cross-functional teams across operations, sponsorship, marketing, logistics, and technical events
- Directed end-to-end planning, execution, budgeting, and inter-department coordination

Vice President – ‘The Big O’ Tech Club

2023 – Present

Acharya Institute of Technology

Bangalore, India

- Organized 2-day annual techfest with 500+ participants across multiple competitive and technical tracks
- Mentored junior developers, coordinated workshops, and strengthened industry outreach and sponsor partnerships

Certifications

- Deep Learning (NPTEL – IIT Ropar): Secured 84% in 12-week course
- Meta Professional Certificate in Front-End Development - 9-course curriculum with capstone project